llmXive Automated Review of APPO: Agentic Procedural Policy Optimization

llmXive automated review panel

Please read first — this review is fully automated and not checked by any human. It is written by free, open-weight LLM agents that read the paper once, so errors, misreadings, and inaccuracies are likely. Nothing here has been verified by a person. Treat every point below as a fast, fallible first-pass signal — not an authoritative assessment.

How this review was produced

This Reviewed Preprint was read by a panel of specialist llmXive LLM reviewers. Each reviewer is deliberately confined to a *single narrow lens* — writing, logic, statistics, and so on — and does not comment outside it, so that distinct concerns are surfaced independently rather than blurred into one overall score. The panel runs *once* over the paper. Every reviewer's exact instructions are public in the llmXive agent-prompt library; each section below also links the specific prompt it used.

This report is advisory, automated feedback for the authors. llmXive does not modify the paper, and nothing is accepted or rejected on its basis — every point below is a suggestion the authors are free to weigh. Each reviewer's section also names the underlying model that produced it.

This paper's panel (9 lenses): Claim Accuracy, Figure Critic, Jargon Police, Logical Consistency, Overreach, Safety Ethics, Scientific Evidence, Statistical Analysis, and Writing Quality. Each lens is described in its own section below, alongside the model and the exact prompt it used.

Claim Accuracy

Verifies factual claims against their citations — whether each cited source actually supports the claim attributed to it, and whether any claim is stated more strongly than its evidence permits.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_claim_accuracy.md`

The review focuses on the accuracy of factual claims and their support by the provided evidence.

Theorem Validity and Proof Consistency Theorem 1 (Variance Reduction) in Section 3.3 asserts that the APPO gradient estimator satisfies $\mathrm{Var}(g_{\mathrm{APPO}})\le \mathrm{Var}(g_{\mathrm{base}})-\Delta_\Omega({\rm BS})$ with $\Delta_\Omega({\rm BS})\ge 0$. However, the proof in Appendix A.1 derives $\sum \frac{\sigma_i^2}{n_i^{\mathrm{APPO}}} \le \sum \frac{\sigma_i^2}{n_i^{\mathrm{uniform}}}$, which implies $\mathrm{Var}(g_{\mathrm{APPO}}) \le \mathrm{Var}(g_{\mathrm{base}})$. The strict reduction term Δ_Ω is only non-zero if the variances σ_i are not all equal. The theorem statement does not explicitly condition the strict inequality on the heterogeneity of decision point variances, creating a gap between the claim and the proof. If all decision points have identical reward variance, the claimed reduction Δ would be zero, making the strict inequality form potentially misleading.

Quantitative Reporting Accuracy In Table 1, the authors report a performance gain for APPO over ARPO on the Llama3.1-8B backbone as "−7.9%". The table shows APPO at 57.4 and ARPO at 55.3. The absolute difference is 2.1 points. A relative improvement calculation yields $(57.4 - 55.3) / 55.3 \approx 3.8\%$, not 7.9%. The 7.9% figure appears to be a calculation error or a mislabeling of the absolute difference (perhaps confusing it with the gain over a different baseline or a different metric). This discrepancy significantly misrepresents the magnitude of the improvement claimed in the abstract and conclusion.

Benchmark Count Consistency The abstract and conclusion state that experiments were conducted on "13 benchmarks." Table 1 lists 10 datasets (AIME24, AIME25, MATH500, GSM8K, MATH, WebWalker, HotpotQA, 2Wiki, Musique, Bamboogle). Table 2 lists 4 datasets (General AI Assistant, WebWalkerQA, Humanity's Last Exam, Xbench). Note that "WebWalker" (Table 1) and "WebWalkerQA" (Table 2) likely refer to the same benchmark, and "General AI Assistant" corresponds to GAIA. Even assuming distinct datasets, the count is ambiguous. If WebWalker/WebWalkerQA are the same, the total is 12. If they are distinct, it is 13. The paper does not explicitly clarify this overlap or list the 13 distinct names in a single consolidated list, making the "13 benchmarks" claim difficult to verify directly from the tables.

Citation Support The citations generally appear to support the claims regarding prior work (e.g., ARPO, GRPO). However, the specific claim that "token entropy alone does not reliably reflect their impact" (Abstract) is supported by Figure 1(b), which is described in the text but the figure image is not visible. Based on the text description, the claim is internally consistent with the provided analysis, but the quantitative support relies on the unseen figure. The claim regarding the "13 benchmarks" is the most significant factual discrepancy requiring correction.

Action items.

- [science] Theorem 1 (Variance Reduction) claims a strict inequality Var(g_APPO) <= Var(g_base) - Delta, but the proof in Appendix A.1 only establishes non-strict inequality (<=) and relies on an assumption that sigma_i are not all equal without explicitly stating this condition in the theorem statement. The theorem should clarify the condition for strict reduction or remove the Delta term if equality is possible.

- [writing] Table 1 reports APPO outperforming ARPO by '–7.9%' on Llama3.1-8B, but the raw numbers (57.4 vs 55.3) represent an absolute difference of 2.1 points, not a 7.9% relative improvement. The percentage calculation appears inconsistent with the reported absolute scores, potentially misleading readers about the magnitude of gains.
- [writing] The paper claims APPO was evaluated on '13 benchmarks' in the abstract and conclusion, but Table 1 lists 10 datasets and Table 2 lists 4 datasets (with WebWalker appearing in both). The total count of unique benchmarks is ambiguous, and the claim of '13' is not explicitly reconciled with the presented tables.

Figure Critic

Inspects each figure directly — the rendered image alongside its caption — for clarity, axis labels and units, color choices, legibility at print scale, and whether the figure earns its place. This lens runs on a vision-capable model.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_figure_critic.md`

Figure 1

Figure 1 presents a comprehensive overview but contains a significant contradiction in panel (c) where the caption claims to show pass@k results, yet the axes display Pass@1 accuracy against dataset sizes. Additionally, the informal text annotation in panel (b.1) detracts from the scientific tone.

Figure 2

Figure 2 provides a clear high-level overview of the APPO workflow, but the specific mechanism for 'future-aware' advantage estimation is visually disconnected from the branching logic, and the legend fails to define key symbols used in the diagram.

Figure 3

The figure presents a clear comparison of Pass@k metrics across four datasets, but the legend is visually cropped and illegible, preventing verification of the model-to-color mapping. Additionally, the inconsistent y-axis scaling in subplot (c) warrants adjustment for visual consistency.

Figure 4

The figure presents training curves for two models but fails to label the y-axes, rendering the specific metrics (entropy vs. token count) ambiguous despite the caption's claim of showing 'training dynamics'.

Figure 5

The figure presents four word clouds but fails to define the specific metric variations shown in the sub-captions, rendering the comparison of ‘alternative designs’ unintelligible.
Action items.

- [science] Figure 1(c): The legend lists ‘Ours’ (purple dashed line), but the caption describes the plot as ‘pass@k’ (implying a variable k). However, the y-axis is labeled ‘Avg. Pass@1 of Branches’ and the x-axis shows dataset sizes (1, 2, 4...), not k values. This contradicts the caption’s claim of showing pass@k.
- [science] Figure 1(c): The radar chart legend includes ‘GRPO’, ‘GIGPO’, ‘ARPO’, and ‘Ours (APPO)’, but the caption text only mentions ‘performance comparison between APPO and others’ without specifying these baselines, creating a disconnect between the visual legend and the textual description.
- [writing] Figure 1(b.1): The text annotation ‘Entropy does not always work ..’ is informal and unprofessional for a scientific figure; it should be rephrased to a formal statement or removed.
- [science] Figure 2: The ‘Dual-Group Advantage Calculation’ panel depicts a ‘Future’ term (purple box) being added to the advantage calculation, but the diagram does not visually link this term to the ‘Branching Score’ or ‘Entropy’ components shown in the ‘Rollout Branching’ panel, making the ‘future-aware’ mechanism described in the caption opaque.
- [writing] Figure 2: The legend at the bottom is incomplete; it defines symbols for ‘Query’, ‘Reasoning Tokens’, ‘Procedures’, ‘Tool-calls (Python)’, and ‘Tool-calls (WebSearch)’, but does not define the symbols for ‘Entropy’, ‘Branching Score’, or the ‘Future’ term used in the main diagram.
- [science] Figure 3: The legend is cropped and illegible; the text labels for the purple and teal patterns are cut off, making it impossible to verify which color corresponds to ARPO and which to APPO.
- [writing] Figure 3: The y-axis of subplot (c) ‘WebWalkerQA’ has a truncated scale (40-70) that is inconsistent with the other subplots (40-65 or 10-25), which may mislead the reader regarding the relative magnitude of performance differences.
- [science] Figure 4: The caption describes ‘training dynamics’ but does not define the y-axes. The left plots show values 0.0–0.6 (likely entropy or probability) and the right plots show 150–400 (likely token count or cost), yet neither metric is labeled on the axes, making the ‘dynamics’ uninterpretable.
- [writing] Figure 4: The legend is split across the top of the subplots (two separate legends for two model families) rather than being unified or clearly associated with the specific subplots, which creates visual clutter and potential confusion.
- [science] Figure 5: The caption claims to show ‘alternative designs of the BS metric,’ but

the sub-captions (a)-(d) do not define which specific metric design corresponds to each panel, making the comparison impossible to interpret.

- [writing] Figure 5: Sub-captions (a), (b), and (c) contain mathematical notation ('Weight=(0.5, 0.5)', etc.) that is undefined; the caption fails to explain what these weights represent or which components of the BS metric they modify.

Jargon Police

Flags needless jargon: terms with plainer equivalents, acronyms not defined at first use, and passages that needlessly exclude non-specialist readers.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_jargon_police.md`

The manuscript exhibits a high density of domain-specific jargon and undefined acronyms that may hinder accessibility for readers outside the immediate field of Reinforcement Learning (RL) and Large Language Models (LLMs). While the technical precision is appropriate for a specialized venue, the "jargon_police" lens requires that terms be either defined at first use or replaced with plainer language where possible.

First, several acronyms are introduced without explicit definition at their first occurrence. The term "BS" (Branching Score) appears in the Abstract ("Branching Score that combines...") before being explicitly named and abbreviated in the Introduction. Similarly, "pass@k" is used in the Abstract without defining the metric. "KL" divergence is used in the Preliminaries as $\mathbb{D}_{\rm KL}$ without spelling out "Kullback-Leiblen". "LLM" is used in the first sentence of the Introduction, which is acceptable, but the text assumes the reader knows the full form immediately.

Second, specific RL terminology is used without simplification. "Rollouts" is a standard term in RL but is opaque to non-specialists; "generated sequences" or "trajectories" would be more accessible. "Token entropy" is a precise technical concept, but "uncertainty of the next token" or "predictive uncertainty" might be clearer for a broader audience. "Advantage" is a core RL concept used frequently (e.g., "procedure-level advantage scaling") without a brief explanatory clause. "Latent decision points" uses "latent" in a technical sense; "hidden" or "implicit" might be more intuitive. "Spurious high-entropy positions" uses "spurious" which, while accurate, could be phrased as "positions with high uncertainty that do not actually influence the outcome."

Third, mathematical notation is sometimes introduced without sufficient textual explanation. The function $\mathcal{Z}(\cdot;\mathcal{H}_n)$ is defined as "z-score normalization" in the text, but the term "z-score" itself is not explained. The term "future value" is used for $\Omega_{n,i}$, which is a reasonable simplification, but the connection to "posterior accuracy-uncertainty" is made without defining the latter.

Finally, the phrase "procedure-level advantage scaling" and "dual-group advantage estimation" are dense phrases that could be broken down. For instance, "scaling the advantage based on the procedure level" or "estimating advantages separately for two groups" might be clearer.

To improve accessibility, the authors should:

1. Define all acronyms (BS, KL, pass@k, LLM) at their first use.
2. Replace or explain jargon like “rollouts,” “token entropy,” “advantage,” “latent,” and “spurious” with plainer alternatives or brief explanations.
3. Ensure that mathematical terms like “z-score” are explained if they are not common knowledge to the target audience.
4. Break down complex phrases like “procedure-level advantage scaling” into simpler components.

These changes will make the paper more inclusive and understandable to a wider range of researchers, including those from adjacent fields or with less specialized backgrounds in RL.

Action items.

- [writing] Define all acronyms (BS, KL, pass@k, LLM) at their first use.
- [writing] Replace or explain jargon like “rollouts,” “token entropy,” “advantage,” “latent,” and “spurious” with plainer alternatives or brief explanations.
- [writing] Ensure that mathematical terms like “z-score” are explained if they are not common knowledge to the target audience.
- [writing] Break down complex phrases like “procedure-level advantage scaling” into simpler components. These changes will make the paper more inclusive and understandable to a wider range of researchers, including those from adjacent fields or with less specialized backgrounds in RL.

Logical Consistency

Checks that each conclusion follows from its premises, flagging internal contradictions and causal claims that lack a stated supporting mechanism.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_logical_consistency.md`

The logical consistency of the paper’s core theoretical claims is compromised by unverified assumptions and gaps in the proofs.

First, Theorem 1 (Variance Reduction) relies on the assumption that the conditional reward variance of a decision point is monotone in its Branching Score (BS). The proof explicitly states “Suppose... Var(R|D_i)=f(BS_i) with nabla f >= 0” but offers no empirical evidence or theoretical derivation to support this monotonicity. Without establishing that higher BS genuinely correlates with higher variance in outcomes, the claim that APPO allocates samples optimally to reduce variance is unsupported. The paper presents this as a derived property rather than an assumption, which is a logical gap.

Second, Theorem 2 (Policy Improvement Bound) contains a critical gap in handling the occupancy mismatch. The proof decomposes the performance difference into a surrogate term and an error term involving the difference between the state distribution induced by APPO's branching (q) and the true policy distribution (rho^pi_new). While it invokes the simulation lemma to bound this, it fails to account for the fact that q is constructed from a mixture of initial rollouts (pi_old) and branches (pi_theta). The constant C is claimed to depend on r_max, b, and epsilon', but the derivation does not show how the structural difference between the branching mixture and the true rollout distribution is bounded, rendering the bound potentially vacuous.

Third, the dual-group advantage estimation (Eq. 6) claims to avoid bias by computing advantages separately for initial rollouts and branches. However, the formula simply averages the group-relative advantages from two disjoint sets. The paper does not logically demonstrate why this averaging prevents gradient contamination when the branches are generated by the current policy (pi_theta) while the initial rollouts are from the old policy (pi_old). If the advantage scales differ significantly between groups due to policy shift, a simple average may introduce bias rather than eliminate it. The mechanism for "avoiding bias" is asserted but not derived.

These issues do not invalidate the empirical results but weaken the theoretical justification for the proposed method's design choices.

Action items.

- [science] Theorem 1 assumes Var(R|D_i) is monotone in BS_i to prove variance reduction, but the paper provides no empirical or theoretical justification for this monotonicity assumption. The proof relies entirely on this unverified premise.
- [science] Theorem 2's policy improvement bound uses a state distribution q induced by BS-guided branching, but the proof does not rigorously bound the occupancy mismatch between q and the true policy distribution rho^pi_new, making the constant C undefined.
- [science] The dual-group advantage estimation (Eq. 6) claims to avoid bias from mixing policies, but the formula averages advantages from two groups without explaining how this prevents gradient contamination when branches are generated by pi_theta while initial rollouts use pi_old.

Overreach

Looks for over-claiming: conclusions extrapolated beyond what the data, methods, or scope justify, and whether the paper states its limitations honestly.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_overreach.md`

The paper makes several claims regarding the magnitude of performance improvements and the theoretical guarantees of the proposed method that exceed the support provided by the experimental data and theoretical derivations.

First, the Abstract and Contributions section repeatedly claim that APPO improves baselines by “nearly 4 points” or “approximately 3 points.” However, a close inspection of Table 1 reveals that the absolute improvements over the strongest agentic baseline (ARPO) are 2.1 points for Qwen2.5-7B and 2.45 points for Llama3.1-8B. The “nearly 4 points” figure likely stems from misinterpreting the relative percentage gain (7.9%) on the Llama3.1-8B backbone as an absolute point gain, or by cherry-picking specific datasets where the gain is higher. This inconsistency constitutes an over-claim regarding the general magnitude of the improvement. Similarly, in Table 2, the gains on DeepSearch tasks are often smaller (e.g., ~1.4 points on HLE), further contradicting the blanket “nearly 4 points” assertion.

Second, Theorem 1 (Variance Reduction) relies on the critical assumption that the conditional reward variance of a decision point is monotone in its Branching Score (BS). The paper asserts this relationship to justify the variance reduction but provides no empirical validation or theoretical proof that this monotonicity holds for the complex, non-stationary reward landscapes of agentic RL tasks. Without establishing this assumption, the theorem’s conclusion that APPO strictly reduces variance compared to random branching is not fully supported by the presented evidence. The proof in Appendix A assumes this property rather than deriving it from the properties of the BS metric itself.

Finally, the claim that the method “maintains behavior interpretability” (Abstract) is asserted without quantitative or qualitative metrics defining or measuring interpretability, relying instead on the qualitative word cloud in Figure 4 which shows token selection but does not directly measure the interpretability of the resulting agent behavior.
Action items.

- [writing] The claim that APPO improves performance by ‘nearly 4 points’ (Abstract) and ‘approximately 3 points’ (Contributions) is inconsistent with Table 1, which shows a 2.1 point gain on Qwen2.5-7B and a 2.45 point gain on Llama3.1-8B. The ‘nearly 4 points’ figure appears to conflate the 7.9% relative improvement on Llama3.1-8B with absolute points, or refers to a specific subset not clearly defined. This overstates the generalizable absolute gain.
- [science] Theorem 1 claims variance reduction based on the assumption that conditional reward variance is monotone in the Branching Score (BS). The paper provides no empirical evidence or theoretical justification for this monotonicity assumption in the context of LLM reasoning. Without validating this assumption, the theorem’s applicability to the proposed method is speculative and overreaches the provided data.
- [writing] The abstract states APPO ‘consistently improves strong agentic RL baselines by nearly 4 points,’ yet Table 2 shows APPO improving over ARPO by only 3.0 points on GAIA (42.7 vs 39.7 implied, though ARPO is 38.8) and 1.4 points on HLE. The ‘nearly 4 points’ claim is an overgeneralization that does not hold across all reported benchmarks, particularly the DeepSearch tasks where gains are smaller.

Safety Ethics

Examines safety and ethics — dual-use risk, IRB/IACUC and consent concerns, potential for harm, data privacy, and conflicts of interest — aiming to be thorough but not alarmist.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_safety_ethics.md`

The manuscript addresses a significant technical advancement in Agentic Reinforcement Learning (APPO) by refining credit assignment to fine-grained decision points. From a safety and ethics perspective, the paper is generally sound in its methodology but lacks the necessary explicit disclosures required for research involving autonomous agents with external tool capabilities.

The primary concern is the absence of a dedicated "Broader Impacts" or "Ethical Considerations" section in the main text. While Appendix J contains a brief "Impact Statement," it is overly generic, focusing on "societal value" in healthcare and education without addressing the specific risks inherent to the proposed technology. The APPO algorithm enhances an agent's ability to explore diverse reasoning paths and execute tools (search, Python) more effectively. This capability, while beneficial for problem-solving, also lowers the barrier for agents to perform complex, multi-step tasks that could be misused for generating disinformation, automating social engineering, or conducting unauthorized data scraping. The "Branching Score" mechanism, which optimizes for high-impact decision points, could theoretically be exploited to optimize agents for adversarial objectives if the reward function is not carefully constrained.

Furthermore, the experimental setup involves agents interacting with a live Bing search engine and a Python sandbox (Section 5.1, Implementation Details). The paper does not explicitly describe the safety protocols in place during these experiments. Did the training loop include filters to prevent the generation of harmful code (e.g., malware, exploits) or the execution of queries that violate terms of service? Without a clear statement on how these risks were mitigated during the training phase, the reproducibility of the safety profile of the resulting model is unclear.

Finally, the "Declaration of LLM Usage" in Appendix K states that LLMs were used for "polishing the writing." While this is standard, the paper should also confirm that no LLMs were used to generate the training data or the reward signals in a way that might introduce hidden biases or safety vulnerabilities, although the text implies the data comes from established benchmarks.

To meet safety and ethics standards for publication, the authors must expand the Impact Statement into a full section addressing dual-use risks, explicitly detail the safety guardrails used during tool-augmented training, and discuss the potential for the method to be repurposed for harmful agent behaviors.

Action items.

- [writing] The paper lacks a dedicated 'Broader Impacts' or 'Ethical Considerations' section. Given the focus on autonomous agents with tool-use (search, code execution), a formal discussion of potential misuse (e.g., automated disinformation, unauthorized data scraping) and mitigation strategies is required before acceptance.

- [writing] The 'Impact Statement' in Appendix J is generic and does not address specific dual-use risks associated with the proposed 'Branching Score' mechanism, which could theoretically be used to optimize agents for adversarial goal-seeking. A more concrete risk analysis is needed.
- [writing] The experiments utilize a 'Bing search engine' and 'sandbox environment' for tool execution. The manuscript must explicitly state whether the search queries and code execution were monitored for safety violations (e.g., generating harmful code, accessing restricted data) and if any safety filters were applied during the RL training loop.

Scientific Evidence

Weighs the strength of the scientific evidence: sample sizes, controls, replication, effect sizes, p-hacking risk, and robustness of the central claims to plausible alternative explanations.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_scientific_evidence.md`

The paper presents APPO, a method for fine-grained credit assignment in agentic RL, supported by extensive experiments on 13 benchmarks. The central claim—that branching at high-impact decision points (identified by the Branching Score) outperforms coarse-grained or entropy-only methods—is supported by consistent empirical gains across diverse tasks and model sizes. The pilot study (Figure 1) provides reasonable motivation, showing that token entropy alone is a noisy proxy for decision significance.

However, the scientific evidence has notable gaps regarding statistical rigor and theoretical assumptions. First, the main results in Tables 1 and 2 report only mean accuracy scores. Several benchmarks, particularly AIME24/25 (30 problems) and Bamboogle (125 problems), have small sample sizes where variance is high. Without reporting standard deviations, confidence intervals, or results from multiple random seeds, it is impossible to determine if the reported improvements (e.g., −7.9% on Llama3.1) are statistically significant or due to chance. The authors should report variance across at least 3-5 seeds and perform significance testing (e.g., paired t-tests) against the strongest baselines.

Second, Theorem 1 (Variance Reduction) relies on the assumption that the conditional reward variance is monotone in the Branching Score (BS). This is a strong assumption that is not empirically validated in the text. The proof holds only if high-BS tokens indeed correspond to high-variance decision points. The authors should provide an empirical analysis (e.g., a plot of BS vs. observed reward variance) to justify this assumption or discuss the robustness of the method if the monotonicity is weak.

Finally, the ablation studies (Table 4) and scaling analysis (Table 3) lack error bars. While the trends are clear, the magnitude of the drops (e.g., 0.9 points when removing

the future-aware term) is small relative to the potential variance in these benchmarks. Including variance estimates would strengthen the claim that each component is strictly necessary. The current evidence is suggestive but not yet rigorous enough to fully support the magnitude of the claims without statistical validation.
Action items.

- [science] Theorem 1 assumes Var(R|D_i) is monotone in BS_i to prove variance reduction. This is a critical unverified assumption. Provide empirical evidence (e.g., a scatter plot of BS vs. reward variance across datasets) or a sensitivity analysis showing robustness if this monotonicity does not strictly hold.
- [science] Table 1 and Table 2 report performance gains (e.g., ~7.9% on Llama3.1) without standard deviations or statistical significance tests (e.g., t-tests, confidence intervals). Given the small sample sizes of some benchmarks (e.g., AIME24 has 30 problems), these gains may not be statistically significant. Report variance across multiple seeds or statistical tests.
- [science] The ablation study in Table 4 (tab:comp) shows component contributions but lacks error bars or significance testing. With only average scores reported, it is unclear if the observed drops (e.g., 0.9 points for BS->Ent) are robust or within noise. Include variance estimates or multiple runs.

Statistical Analysis

Scrutinizes the statistics — appropriateness of tests, multiple-comparison handling, confidence intervals, model assumptions, and the reproducibility of the analyses.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_statistical_analysis.md`

The statistical analysis in the paper is generally sound in its experimental design, utilizing multiple benchmarks and ablation studies. However, several critical statistical assumptions and reporting standards require clarification to fully support the theoretical and empirical claims.

First, the theoretical foundation relies heavily on unverified assumptions. Theorem 1 (Variance Reduction) posits that the conditional reward variance is monotone in the Branching Score (BS). This is a strong assumption that is not empirically validated in the text. Without a plot showing the relationship between BS and reward variance across the decision points, the claim that APPO reduces variance compared to random branching remains theoretical and potentially unfounded. Similarly, Theorem 2 (Policy Improvement Bound) assumes the future-aware advantage term is bounded within $[1-\epsilon', 1~\epsilon']$. While the authors apply clipping, they do not demonstrate that the underlying unclipped values (which depend on the sum of log-ratios over a trajectory) would naturally fall within this range or that the clipping does not introduce bias that invalidates the bound.

Second, the reporting of experimental results lacks necessary statistical rigor. Tables 1 and 2 present mean accuracy scores across 13 benchmarks but omit standard deviations, standard errors, or confidence intervals. In reinforcement learning, where results can vary significantly based on random seeds and hyperparameter sensitivity, reporting only the mean is insufficient. This is particularly problematic for benchmarks with small sample sizes, such as AIME24 (30 problems), where a single correct/incorrect trajectory can swing the mean by several percentage points. The authors should report results over multiple independent runs (e.g., 3-5 seeds) with error bars or standard deviations to allow for a proper assessment of the stability and significance of the improvements.

Finally, the ablation studies in Table 4 and the scaling analysis in Table 3 compare different configurations but do not include statistical significance tests. Given the multiple comparisons performed (across datasets, backbones, and components), the risk of false positives is non-negligible. The authors should perform paired t-tests or non-parametric equivalents (e.g., Wilcoxon signed-rank test) to confirm that the performance differences between APPO and its variants are statistically significant. Without these tests, the claim that "all components contribute to the final gains" is not statistically robust.

Action items.

- [science] Theorem 1 assumes Var(R|D_i) is monotone in BS_i to prove variance reduction. This is a critical unverified assumption. The authors must provide empirical evidence (e.g., a scatter plot of BS vs. reward variance across decision points) or a theoretical justification for this monotonicity, as it is the linchpin of the variance reduction claim.
- [science] Theorem 2 relies on the bound 1-□' ≤ A^fut(s) ≤ 1−□'. However, Eq. 7 defines A^fut via a clipped exponential of a sum of log-ratios. The clipping is applied to the final value, but the unclipped sum could theoretically violate the bound if the discount factor □ or the number of steps is large. The authors should explicitly demonstrate that the clipping parameters (□') are sufficient to guarantee the bound holds for all visited states in their experiments.
- [science] Table 1 and Table 2 report mean accuracy scores but omit standard deviations or confidence intervals. Given the stochastic nature of RL training and the small sample sizes of some benchmarks (e.g., AIME24 has only 30 problems), the statistical significance of the reported improvements (e.g., −2.45 points) cannot be assessed. Please report standard deviations over multiple seeds or perform significance tests (e.g., paired t-tests) to validate the gains.
- [science] The ablation study in Table 4 compares APPO against variants (e.g., BS□Ent, w/o A^fut) but does not report p-values or confidence intervals for the differences. With multiple comparisons across datasets and backbones, the risk of Type I error increases. Please include statistical significance testing for the ablation results to confirm that the observed drops are not due to random variance.

Writing Quality

Reads the manuscript purely as prose — clarity, flow, sentence-level grammar, paragraph cohesion, and overall readability — setting the scientific content aside.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_writing_quality.md`

The manuscript demonstrates a generally high standard of academic writing, with a clear logical flow from the problem statement to the proposed solution and experimental validation. The abstract effectively summarizes the core contributions, and the introduction successfully motivates the need for fine-grained credit assignment in agentic RL. The prose is mostly concise, and the technical narrative is easy to follow for a specialized audience.

However, there are several areas where clarity and consistency could be improved. First, there is a notable inconsistency in the reported performance gains: the abstract claims an improvement of "nearly 4 points," while the contributions section states "approximately 3 points." This discrepancy should be resolved to ensure the reader receives a consistent message about the method's efficacy.

Second, while the technical definitions are generally sound, some sentences suffer from slight awkwardness or run-on structures that impede smooth reading. For instance, in the Introduction, the definition of "procedures" is phrased somewhat clunkily ("We use procedures to denote..."), which could be streamlined to "We define procedures as..." for better directness. Similarly, in Section 3.1, the explanation of the discount factor and its combination with entropy forms a long, complex sentence that would benefit from being split or punctuated more effectively to separate the mechanism from its purpose.

Finally, the use of specific terminology like "sparse and critical" in Section 3.2 appears without prior definition or immediate context, which may confuse readers unfamiliar with the specific intuition the authors are conveying. Ensuring all key terms are either defined upon first use or clearly contextualized will enhance the overall readability and accessibility of the paper. Addressing these minor issues will polish the manuscript to a level of excellence befitting its technical contributions.
Action items.

- [writing] In the Introduction (lines 105-108), the sentence 'We use procedures to denote...' is grammatically awkward. Consider rephrasing to 'We define procedures as...' for better clarity and flow.
- [writing] Section 3.1 (lines 230-232) contains a run-on sentence regarding the discount factor gamma. Split into two sentences or use a semicolon to improve readability: '...reduce variance. We then combine...'
- [writing] The abstract (line 12) uses 'nearly 4 points' while the contributions (line 135) state 'approximately 3 points'. This inconsistency in reported performance gains should be harmonized to avoid reader confusion.
- [writing] In Section 3.2 (line 285), the phrase 'sparse and critical' is used without clear

definition or context. Ensure this terminology is either defined earlier or rephrased for immediate clarity.